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PROFESSIONAL SUMMARY:

- Around 5 years of experience in Data Analysis and Modeling, catering to client business needs and delivering efficient solutions within committed timelines.
- Proficient in Data Analysis, Cleansing, Transformation, Migration, Integration, Import, and Export using ETL tools, IDQ primarily Informatica.
- Extensive experience in SAS and SQL Coding, data modeling, and data mining.
- Hands-on learning with different ETL tools to prepare data for Tableau through Tableau Data Extract.
- Expertise in writing complex SQL queries, utilizing Indexing, Aggregation, and materialized views for query optimization.
- Leverage GCP services such as BigQuery, Dataflow, Pub/Sub, Dataproc, and Data Studio for data modeling and analytics.
- Demonstrating knowledge of streaming data concepts and understanding the fundamentals of event-driven architectures.
- Experience with Kafka for real-time processing and analysis of data streams.
- Hands-on experience in complex query writing and query optimization for Oracle, T-SQL, and SQL Server.
- Translating business requirements into data science and technical requirements.
- Solid understanding of Data Modeling, Data Collection, Data Cleansing, Data Warehouse/Data Mart Design, ETL, BI, OLAP, and Client/Server applications.
- Driving cross-functional analytics projects from inception to machine learning, visualization, and presentation.
- Expertise in Business Intelligence, Data Warehousing, ETL, and Big Data technologies.
- Experience creating ETL mappings with Informatica for moving data from various sources to Data Warehouse.
- Writing PL/SQL statements for Stored Procedures, Functions, Triggers, and packages.
- Skilled in Tableau Desktop versions 10x for data visualization, reporting, and analysis.
- Developing reports, dashboards using Tableau for quick reviews presented to Business and IT users.
- Performing predictive Modeling, Pattern Discovery, Market Basket Analysis, Segmentation Analysis, Regression Models, and Clustering.
- Capture metadata attributes such as data source, data owner, data quality, data usage, and access controls to facilitate data governance and stewardship.
- Hands on experience in Amazon Web Services AWS provisioning and good knowledge of AWS services like EC2, S3, Red shift, Glacier, Bamboo, ELB (Load Balancers), RDS, SNS, SWF and EBS etc. and Azure
- A Property and Casualty (P&C) data analyst should possess strong data analysis skills, including the ability to gather, clean, and analyze large datasets specific to property and casualty insurance.
- Cloudera certified developer for Apache Hadoop, with good knowledge of Cassandra, Hive, Pig, HDFS, Sqoop, and MapReduce.
- Adept in designing visualizations using SAS and Tableau, Dashboards, Business Metrics, Operating Statistics, and Graphs.
- Develop and maintain a centralized metadata repository or data catalog to document data assets, schemas, definitions, and lineage.
- Developed and automated data workflows using Databricks Notebooks, improving efficiency and scalability of analytics processes.
- Proficient in Experimental Design, Sampling, Linear and Logistical Regression, and Decision Trees.
- Experience working with Amazon Marketing Cloud platform and other ECommerce analytics tools.
- Expert in Data Analysis, identifying Datasets, Source Data, Source Metadata, Data Definitions, and Data Formats for insurance data.
- Acquired knowledge on Big Data technologies, mainly Hadoop, Spark, Hive, Pig, and Tableau frameworks.
- Working with Amazon Web Services (AWS) for applications focusing on high-availability, fault tolerance, and auto-scaling.
- Participated in developing research methodologies for clinical data in the insurance domain.
- Published workbooks and data sources to Tableau Server for further review and analysis.
- Experience in storing and managing large datasets in S3 buckets.
- Designed, developed, and implemented Workload Analysis for Medical Secretary Role in the insurance context.
- Implementing strategies for managing data lifecycle, versioning, and access control within S3.
- Excellent communicative, interpersonal, intuitive, analysis, leadership skills, quick starter with the ability to master and apply new concepts

TECHNICAL SKILLS:

Programming Languages	R Programming, Python, MatLab, VB, Java, C, C++, SQL, MySQL, PL/SQL
ETL Tools	Informatica Power Center 9.1/8.6 (Designer, Workflow Manager/Monitor, Repository), Ab Initio
Testing Tools	Jira, HP ALM, IBM ClearQuest, IBMRQM, MTM, SDLC
Database Tools	Oracle SQL Developer, Toad, Oracle 10g/11g/12c, MS SQL Server, SSIS, SSRS, Data Grid
Big Data Technologies	Cassandra, Pig, Hive, HDFS, Map Reduce, Sqoop, Yarn
BI and Analytics Tools	OBIEE, Oracle Reports Builder, Spotfire, Tableau 10.5, Pandas, Seaborn, Matplotlib, Cognos, Excel, SAS, SAS Enterprise Miner
Data Modeling	Regression Modeling, Time Series Modeling, PDE Modeling, Star-Schema Modeling, Snowflake-Schema Modeling, FACT and Dimension tables
Statistical Modeling/Algorithms	Regression Modeling, Time Series Analysis, ACT, Random Forest
Cloud Tools	AWS, Azure, Google Cloud Platform, Google Big Query, Github

PROFESSIONAL EXPERIENCE:**Client: Southwest Airlines, Dallas, TX****Dec 2023 – Till Date****Role: Data Analyst****Responsibilities:**

- Participated in all phases of Data Mining, Data Collection, Data Modelling, Data Cleaning, Developing-Models, Validation, Visualization and Performed Gap Analysis.
- Used Pandas, NumPy, Seaborn, SciPy, and Matplotlib & Sci-kit-learn in Python for developing various Machine Learning Algorithms.
- Analyzed and documented business requirements, both functional and non-functional, for enhancing the existing CRM system
- Created application prototypes and screen mockups to visualize proposed system enhancements and validate with stakeholders.
- Conducted comprehensive Gap Analysis for large-scale applications, identifying discrepancies between current (As-Is) and desired future state (To-Be) to drive process improvements and system enhancements.
- Created and maintained comprehensive documentation including business requirements, functional specifications, process flows, use cases, and user stories, incorporating Gap Analysis findings to clearly articulate areas of improvement.
- Collaborated with cross-functional teams to design and implement SQL-based reporting solutions and dashboards for monitoring key performance indicators (KPIs) with the internal data.
- Developed and implemented robust data quality rules and monitoring processes to ensure data accuracy and integrity across financial and operational data sets.
- Developed and implemented predictive models using machine learning algorithms such as Linear Regression, Classification, Multivariate Regression, Naive Bayes, Random Forest, K-means Clustering,
- Developed graphical representations of complex business processes using Microsoft Visio.
- Documented and implemented data transformation rules and data quality monitoring processes, ensuring accurate and consistent data flow. Developed predictive models and ensured quality standards across multiple data projects using Python and SQL
- Used data modeling tools such as ERwin and PowerDesigner to define data relationships and structures for large-scale applications.
- Utilized Microsoft products such as Visio, PowerPoint, Excel, and Word to create project documentation and presentations.
- Setup database in AWS using RDS and configuring backups for S3 bucket.
- Worked on Amazon EC2 Clusters to deploy files into Buckets.
- Implemented various data pre-processing techniques to manipulate the unstructured, structured data, data imputations and data imbalance issues.
- Mapped complex business processes to data models to streamline data integration efforts within AWS Cloud environments, ensuring alignment with business requirements.
- Written complex SQL queries to extract, manipulate, and analyze data from relational database such as MySQL.
- Integrate Statistical Reports created in Python/R with Tableau to create dashboards and generate varied reports in accordance with the situational demands.
- Documented training and dashboard design best practices and trained clients with reporting feature.
- Experience in gathering, cleaning, and analyzing financial data using Bloomberg tools to identify trends, patterns, and insights for informed decision-making.
- Utilized Spark, Spark Streaming, MLlib, Python, a broad variety of machine learning methods including Classifications, Regressions, Clustering, dimensionally reduction etc.
- Utilized Databricks and Spark SQL for real-time data comparisons and processing between Snowflake and other data sources, maintaining data accuracy and consistency.
- Used Spark Data frames, Spark-SQL, Spark MLlib extensively and developing and designing POC's using Python, Spark SQL and MLlib libraries.
- Strong understanding of banking regulations, financial markets, and investment strategies related to treasury functions.
- Participated in the design and development of data Lakehouse architecture, incorporating best practices for data governance and data quality monitoring.
- Expert in using Business Intelligence and Data Visualization tool Tableau.
- Utilized MS SQL, data warehousing programs, Tableau deciding useful metrics for performance measurement and other dashboard/visualization toolsets/libraries for data intelligence and analysis.
- Conducted data profiling and quality checks within Databricks to ensure the integrity of datasets across cloud-based platforms.
- Processed and normalized data from various bespoke sources, converting them into standardized formats to streamline data integration and analysis.
- Perform data comparison between SDP (Streaming Data Platform) real time data with AWS S3 data and Snowflake data using Data bricks, Spark SQL, and Python.

Environment: Python 3.x (Scikit-learn, SciPy NumPy, Pandas, NLTK, Matplotlib, Seaborn), Machine Learning (Logistic regression/ Random Forests/ KNN/ K-Means Clustering / Hierarchical Clustering/ Ensemble methods), Erwin, informatica IDQ, Scala, Deep Learning (CNN,RNN, LSTM), NLP, Big Data, Apache Spark/PySpark/MLlib/Spark-SQL, Kafka, Zookeeper, AWS, PostgreSQL, Tableau, Agile

Client: Travelport , Denver, CO**Sep 2022 – Nov 2023****Role: Data Analyst****Responsibilities:**

- Gathered data and business requirements from end-users and management.
- Designed and built data solutions to migrate existing source data in Teradata and DB2 to Big Query on the Google Cloud Platform.
- Designed Scoop scripts to load data from Teradata and DB2 to the Hadoop environment.
- Developed Shell scripts to transfer data from Hadoop to Google Cloud Storage (GCS) and from GCS to Big Query.
- Builds mappings and workflows in IDQ workbench by taking advantage of the Effort Administrator's service connections.

- Utilized MongoDB's query operators, aggregation pipeline stages, and indexing strategies to optimize query performance and data retrieval.
- Designed data models and document schemas for NoSQL databases based on application requirements and data access patterns.
- Utilized ETL tools such as Informatica to integrate and transform data from Teradata and DB2 into BigQuery, ensuring accurate data migration and normalization.
- Extracted, cleaned, and analyzed large datasets from Amazon Marketing Cloud platform to identify trends, patterns, and opportunities for optimization.
- Led the development of Power BI dashboards, including cross-maps, scatter plots, and heatmaps, showcasing trends, forecasts, and regressions.
- Designed and implemented data integration pipelines in Azure Data Factory to ingest, transform, and load (ETL) data from various source systems into Azure data lakes or data warehouses.
- Designed and implemented ETL pipelines in Databricks for large-scale data processing, ensuring seamless integration of structured and unstructured datasets.
- Continuously evaluate and enhance the functionality, usability, and effectiveness of the data catalog based on user feedback, industry trends, and evolving business needs.
- Interpret and communicate findings to non-technical stakeholders in a clear and actionable manner.
- Utilized Snowflake's virtual warehouse architecture to scale compute resources dynamically, ensuring high performance and concurrency for data processing workloads.
- Experience with AWS Glue for building and managing ETL pipelines to process and transform data from various sources into a usable format.
- Implemented data quality solutions using Informatica Data Quality (IDQ), improving data accuracy and consistency across multiple projects.
- Create visually appealing and easy-to-understand dashboards using tools like Tableau or Power BI.
- Developed Tableau Data Visualizations, including Cross Map, Scatter Plots, Geographic Map, and Heatmaps.
- Implemented analyzed data into Tableau, showing regression, trend, and forecast in dashboards.
- Stay abreast of emerging trends and technologies in data modeling, cloud computing, and GCP services.
- Involved in back-end testing and addressed data quality issues.
- Created source-to-target data mappings, business rules, and business/data definitions.
- Collaborated with ETL/Informatica IDQ teams to source data and performed data analysis to identify gaps.
- Integrated Azure Data Factory with source systems and data stores including Azure Blob Storage, Azure SQL Database, Azure Synapse Analytics (formerly SQL Data Warehouse), and on-premises databases.
- Designed data marts using dimensional data modeling with star and snowflake schemas.
- Created interactive dashboards and reports using tools like Tableau to visualize key eCommerce metrics and monitor performance.
- Apply security best practices for data stored and processed in GCP, including encryption, access controls, and compliance standards.
- Documented data transformation processes and configurations, providing clear guidelines for normalization and integration tasks.
- Integrating Kafka with various data sources to capture and stream data in real-time.
- Setting up SNS to receive notifications on data-related events, such as the completion of ETL jobs or the arrival of new data in S3.
- Provide training and documentation to users on how to navigate, search, and leverage the data catalog for data discovery, analysis, and reporting purposes.
- Demonstrating the ability to architect end-to-end data flows, starting from data ingestion through S3, processing with Glue and Lambda, querying with Athena, and notifying relevant stakeholders with SNS.
- Implementing cost-effective solutions by optimizing data storage, processing, and resource utilization across these AWS services.

Environment: Oracle 12c, T-SQL, SQL Server, SQL, GCP, Amazon Redshift, Snowflake, Power BI, MongoDB, PostgreSQL Big Query, SAS, Star Schema, Tableau and Python, Amazon EMR, Lambda, Databricks, Data Migration, Data Warehouse, Microsoft Azure, Cassandra, NoSQL, Hadoop, Git, GitHub, Teradata, Linux, Spark, Docker, Kubernetes, PySpark.

Client: Meta Minds, Hyderabad, India

Jan 2021 – July 2022

Role: Data Analyst

Responsibilities:

- Worked with Data Analysis Primarily Identifying Data Sets, Source Data, Source Metadata, Data Definitions Data Formats, Data Validation, Data Cleansing, Data Verification, and Identifying Data Mismatch.
- Involved in Data Architecture Designing and development of a Central Data Repository of Static Content for Syndicated Loans business. Liaison with the Syndicated Loans team as well as downstream clients in the bank to define scope and requirements.
- Designed a STAR schema for the detailed data marks and Plan data marks involving confirmed dimensions.
- Involved in the Data Integration and Infrastructure project for the Valley bank. We assess the impact on downstream data management environments of processing different banking products through different systems and processes.
- Performed Count Validation, Dimensional Analysis, Statistical Analysis and Data Quality Validation in Data Migration.
- Generated portfolio analysis and P&L reports by performing risk analysis and risk management (Credit Risk, Market Risk, Reinvestment Risk, and Inflation Risk) based on the client investment objectives.
- Designed and implemented multiple dashboards using Power BI - PowerPivot & Power Query tools for in-house metrics.
- Simulated the trading strategies using historical data to assess potential performance and refine strategies.
- Collaborated with the traders and quants to formulate trading strategies based on data insights.
- Used the AWS -CLI to suspend on Aws Lambda function used AWS CLI to automate backup of ephemeral data stores to S3 buckets EBS.
- Involved in translating the business requirements into data requirements across different systems.
- Involved in understanding the customer needs with regards to data, documenting requirements and complex SQL statements to extract the data and packaging data for delivery to customers.
- Used AWS for application deployment and configuration.

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- Wrote complex SQL and PL/SQL queries to identify granularity issues and relationships between data sets and created recommended solutions based on analysis of the query results.
- Performed unit testing on transformation rules to ensure data moved correctly.
- Maintained Excel workbooks, such as development of pivot tables, exporting data from external SQL databases, producing reports and updating spreadsheet information.
- Researched and fixed data issues pointed out by QA team during regression tests.
- Participates in the development of Enhancement for the current Commercial and Mortgage Securities.
- Manipulate and prepare data, extract data from database for business analyst using Tableau.
- Review normalized schemas for effective and optimum performance tuning queries and data validations in OLTP and OLAP environments.
- Developed and implemented data integration solutions using Talend to streamline the extraction, transformation, and loading (ETL) processes.
- Employed Talend for data cleansing, normalization, and transformation, ensuring data quality and consistency across multiple datasets.
- Developed dashboards and ad-hoc reports using MS Power BI for senior management team for analysis.
- Wrote calculated columns, Measures query's using DAX in Power BI to show good data analysis techniques.
- Involved in scorecard and KPI Reporting.
- Created different tabular reports using Power BI features and enhanced them based on user requirements.
- Utilized Power BI to design multiple scorecards and dashboards to display information required by different departments and upper-level management.

Environment: MS SQL Server, Microsoft Management Studio, Microsoft Visual Studio, Google Big Query, RapidMiner, MSSQL Server Integration Services (SSIS), SQL Server Reporting Services (SSRS), SQL, Power BI, Tableau, Python, AWS.

Client: Techmagix IT Pvt Ltd, India

Sep 2019 – Dec 2020

Role: Business Data Analyst

Responsibilities:

- Performed Exploratory Data Analysis, trying to find trends and clusters.
- Built models using techniques like Regression, Tree based ensemble methods, Time Series forecasting, KNN, Clustering and Isolation Forest methods.
- Worked on data that was a combination of unstructured and structured data from multiple sources and automated the cleaning using Python scripts.
- Communicated and coordinated with other departments to collect business requirements.
- Tackled highly imbalanced Fraud dataset using under sampling with ensemble methods, oversampling and cost sensitive algorithms.
- Improved fraud prediction performance by using random forest and gradient boosting for feature selection with Python Scikit-learn.
- Built scalable and deployable machine learning models.
- Utilized Sqoop to ingest real-time data. Used analytics libraries Sci-Kit Learn, MLLIB and MLX tend.
- Extensively used Python's multiple data science packages like Pandas, NumPy, and matplotlib, Seaborn, SciPy, Scikit-learn and NLTK.
- Collaborated with testing teams to ensure that developed solutions meet business requirements and quality standards through the creation and execution of test plans and test cases, utilizing Gap Analysis findings to develop comprehensive test cases that address identified gaps and ensure the desired future state is achieved.
- Implemented machine learning model (logistic regression, XGBoost) with Python Scikit-learn.
- Optimized algorithm with stochastic gradient descent algorithm Fine-tuned the algorithm parameter with manual tuning and automated tuning such as Bayesian Optimization.
- Developed a technical brief based on the business brief. This contains detailed steps and stages of developing and delivering the project including timelines.
- After signing off from the client on technical brief, started developing the SAS codes.
- Wrote the data validation SAS codes with the help of Univariate, Frequency procedures.
- Summarizing the data at customer level by joining the datasets of customer transaction, dimension and from 3rd party sources.
- Separately calculated the KPIs for Target and Mass campaigns at pre-promo-post periods with respect to their transactions, spend and visits.
- Measured the ROI based on the difference's pre-promo-post KPIs.
- Worked extensively with data governance team to maintain data models, Metadata and dictionaries.
- Used Python to preprocess data and attempt to find insights.
- Iteratively rebuild models dealing with changes in data and refining them over time.
- Created and published multiple dashboards and reports using Tableau server.
- Extensively used SQL queries for legacy data retrieval jobs.
- Tasked with migrating the Django database from MySQL to PostgreSQL.
- Gained expertise in Data Visualization using matplotlib, Bokeh and Plotly.
- Responsible for maintaining and analyzing large datasets used to analyze risk by domain experts.
- Developed Hive queries that compared new incoming data against historic data. Built tables in Hive to store large volumes of data.
- Used Big data tools Spark (Spark SQL, MLlib) to conduct the real time analysis of credit card fraud based on AWS *
- Performed Data audit, QA of SAS code/projects and sense check of results.

Environment: Spark, Hadoop, AWS, SAS Enterprise Guide, SAS/MACROS, SAS/ACCESS, SAS/STAT, SAS/SQL, ORACLE, MS-OFFICE, Python (scikit-learn, pandas, Numpy), Machine Learning (logistic regression, XGBoost), Gradient Descent algorithm, Bayesian optimization, Tableau.